

10 One and two sample non-parametric tests

Subject content	Additional information
10.1 Use sign or Wilcoxon signed-rank tests to investigate population median in single sample tests and also to investigate for differences using a paired model.	Students will be expected to use the sign test or the Wilcoxon signed-rank test to: • test the value of a single population median based on a single sample • test for a difference between two population medians based on a sample of 'paired' data. Students will be expected to learn and recall necessary conditions for the validity of these tests, and may be required to use these to select the test that is most appropriate in a given context. Sign test – no assumptions necessary regarding the distribution for test to be valid. Wilcoxon signed-rank test – assumption that the distribution is symmetrical, or that the distribution of differences is symmetrical in the case of a paired test, for test to be valid.
10.2 Use the Wilcoxon rank-sum test to investigate for difference between independent samples.	Students will be expected to test for a difference between two population medians, based on two independent samples, using the Wilcoxon rank-sum test (this is also known as the Mann-Whitney test).